EXPLORATORY DATA ANALYSIS USING POWER BI

REPORT
WEEK - 4

Shopify Financial Performance Dashboard - DAX Measures and Visualizations

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Introduction:

The Shopify Financial Performance Dashboard project uses Power BI to analyze and visualize historical Shopify stock data. The dashboard presents stock prices, trading volume, financial trends, and market performance using interactive charts and DAX measures. It helps users understand changes in Shopify's stock performance and identify important trends from historical financial data.

Objective:

The main objectives of the Shopify Financial Performance Dashboard are:

1. To analyze historical Shopify stock data using Power BI.

2. To visualize stock price trends using Open, High, Low, and Close price data.

3. To calculate 20-day and 50-day Simple Moving Averages using DAX.

4. To analyze trading volume and compare it with moving averages.

5. To identify bullish and bearish trends using moving average comparisons.

6. To compare stock performance across different time periods.

7. To provide interactive date filtering using slicers.

8. To enable dynamic metric selection for different stock price values.

9. To provide detailed financial information through custom tooltips.

10. To present complex financial data in a simple and understandable dashboard.

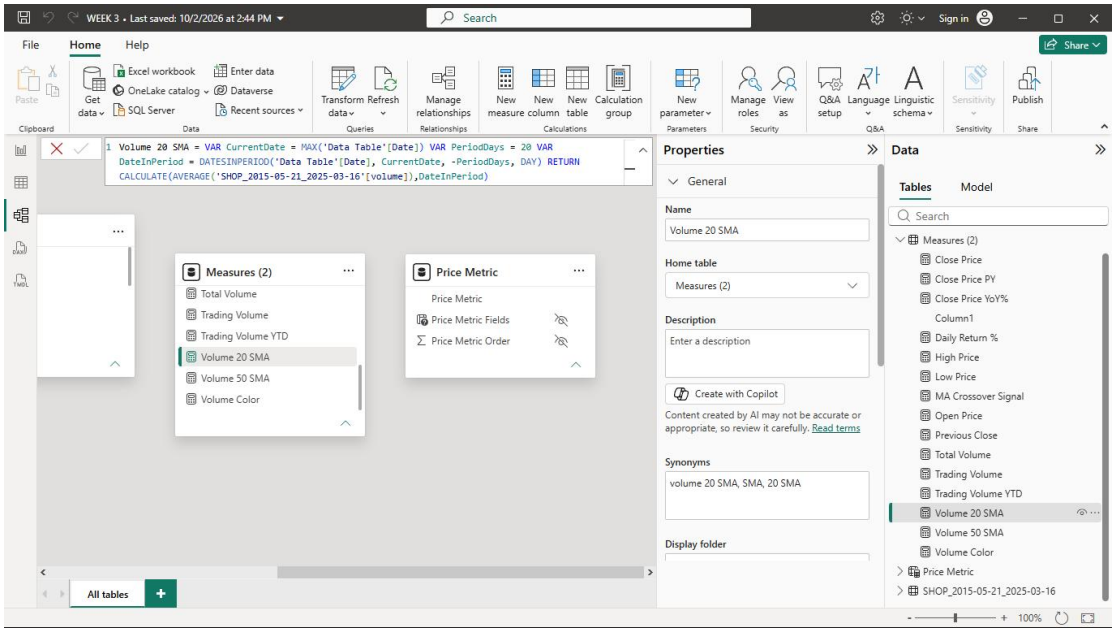
Steps Involved:

1.)Create 20-Day Moving Average measure using DAX.

The 20-day volume SMA calculates the average trading volume over the latest 20 days.

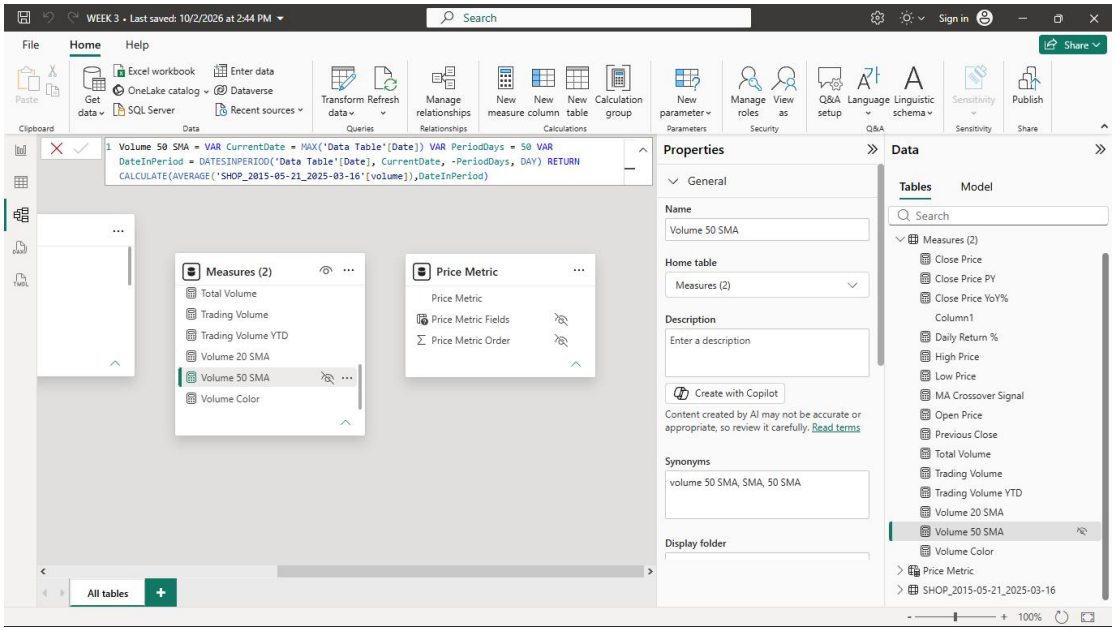
```
Volume 20 SMA =
VAR CurrentDate = MAX('Date Table'[Date])
VAR PeriodDays = 20
VAR DateInPeriod =
 DATESINPERIOD('Date Table'[Date],
 CurrentDate,
 -PeriodDays,
DAY)
RETURN CALCULATE(
 AVERAGE('SHOP_2015-05-21_2025-03-16'[volume]),
```

DateInPeriod
)



2.)Create 50-Day Moving Average measures using DAX.

```
Volume 50 SMA =
 VAR CurrentDate = MAX('Date Table'[Date])
 VAR PeriodDays = 50
 VAR DateInPeriod = DATESINPERIOD('Date Table'[Date],
 CurrentDate,
 -PeriodDays,
DAY)
RETURN CALCULATE(
AVERAGE('SHOP_2015-05-21_2025-03-16'[volume]),
DateInPeriod
)
```

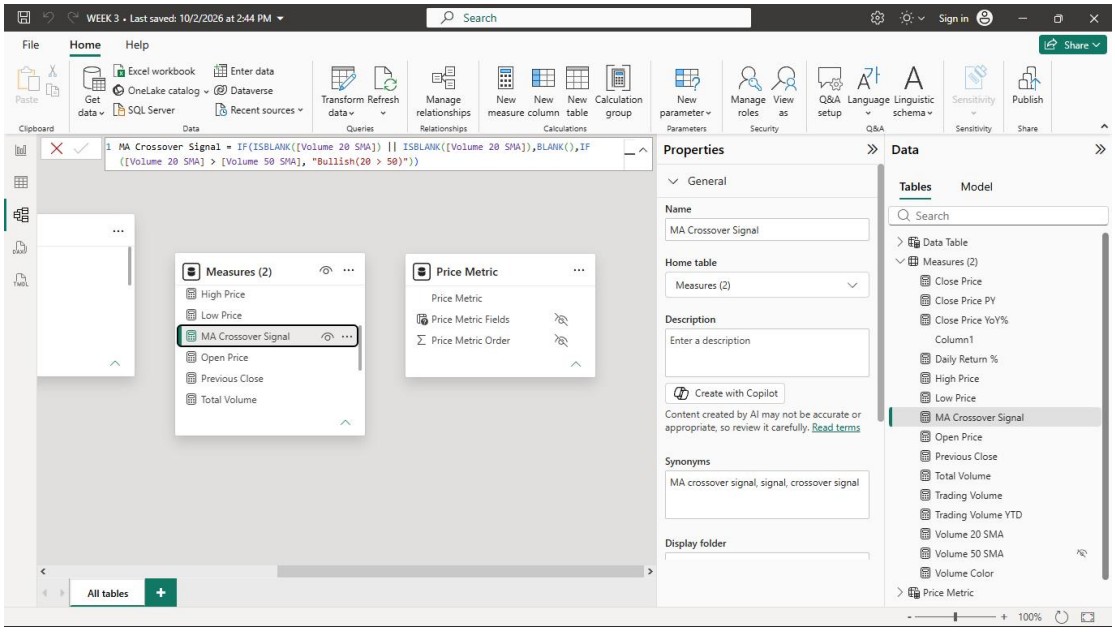


3.)Develop a Moving Average measure.

The crossover signal compares the 20-day SMA with the 50-day SMA.

```
MA Crossover Signal = IF(
ISBLANK([Volume 20 SMA]) || ISBLANK([Volume 50 SMA]),
BLANK(),IF([Volume 20 SMA] > [Volume 50 SMA],
"Bullish(20 > 50)" ,
"Bearish(20 < 50)"
 )
)
```

This measure helps identify whether the short-term moving average is above or below the long-term moving average.

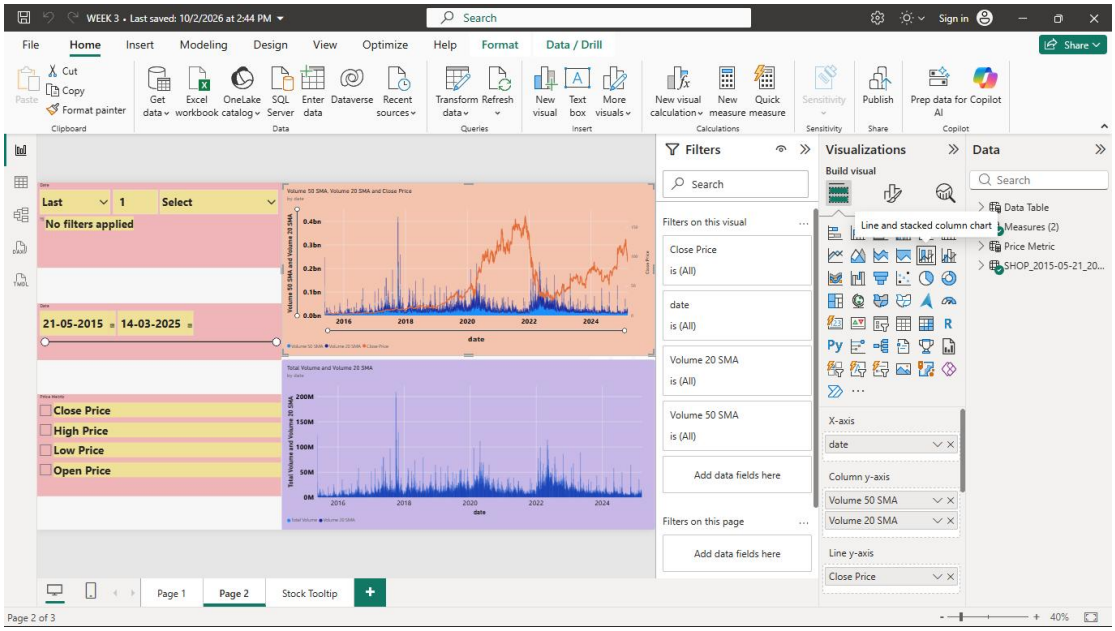


Creating Visualizations in Power BI:

After creating the DAX measures, the next step is to create interactive visualizations.

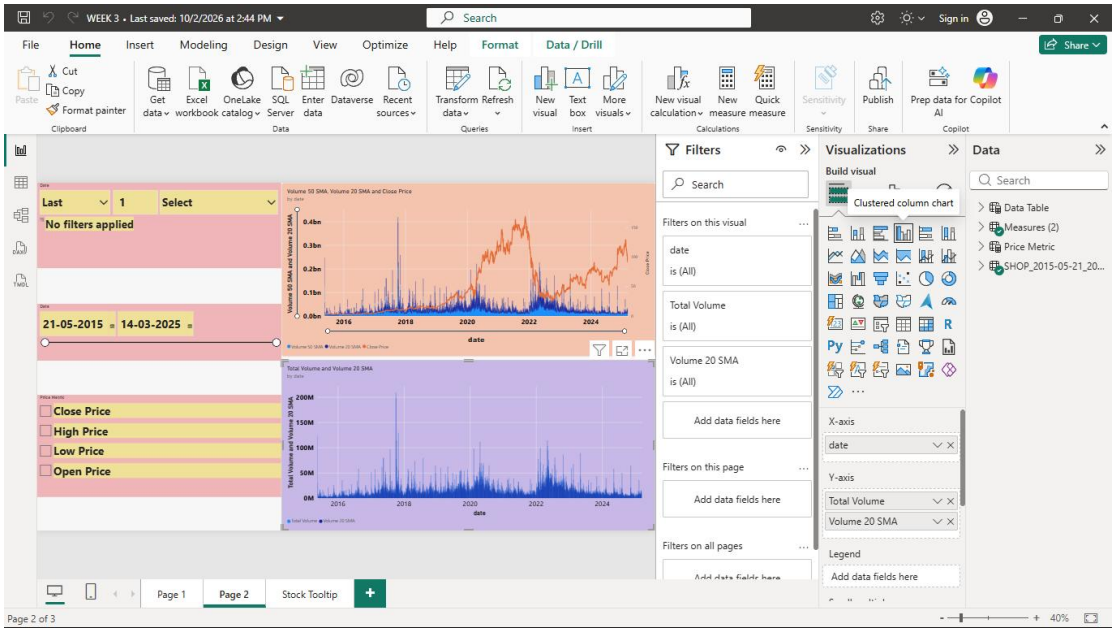
1.)Price Trend Chart:

Select Line and Stacked Column Chart from the visualizations pane and build an interactive price trend chart containing Close Price, 20 SMA AND 50 SMA.



2.)Trading Volume Visualization:

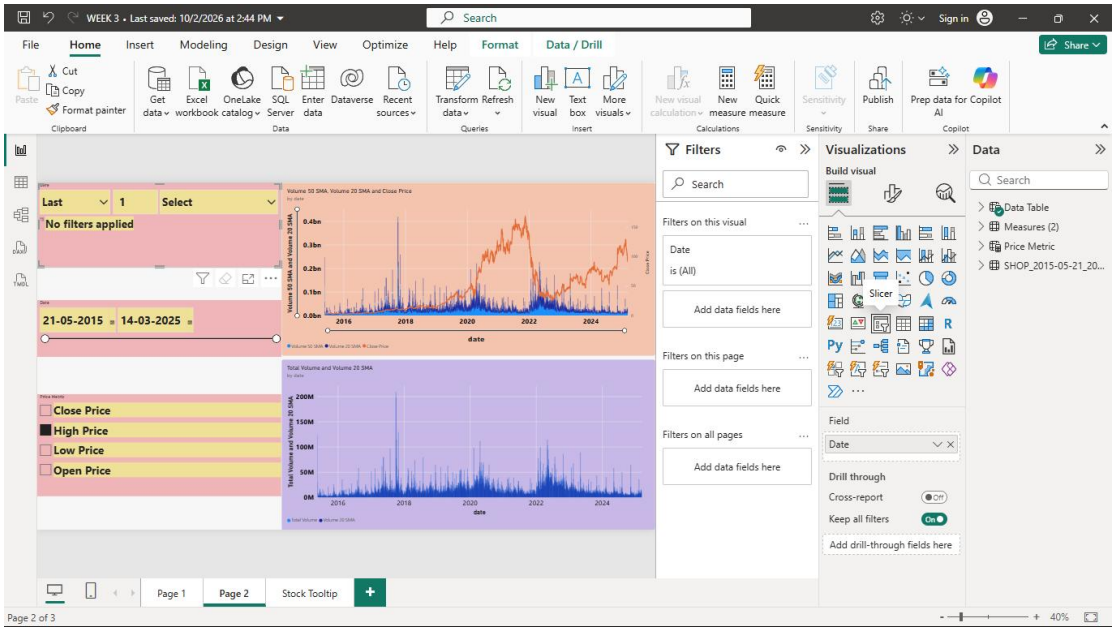
Select a Clustered Column Chart and create a trading volume visualization using Date, Total Volume and Volume 20 SMA.



3.)Timeline Slicer:

Add a Slicer visual.
Drag: Date Table[Date] into the slicer.

Set the slicer type to Between. This allows the user to select a specific start date and end date
and dynamically filter the stock analysis.

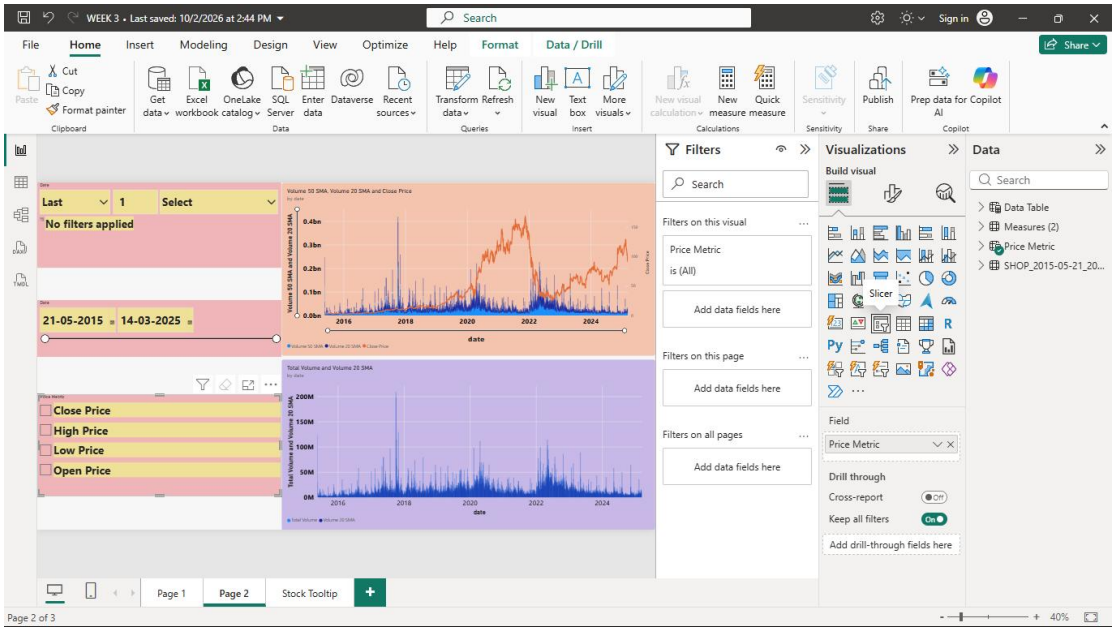


4.) Metric Selector:

To allow users to switch between different price metrics:
Go to:
Modeling → New Parameter → Fields
Create a parameter called:
Price Metric
Add:

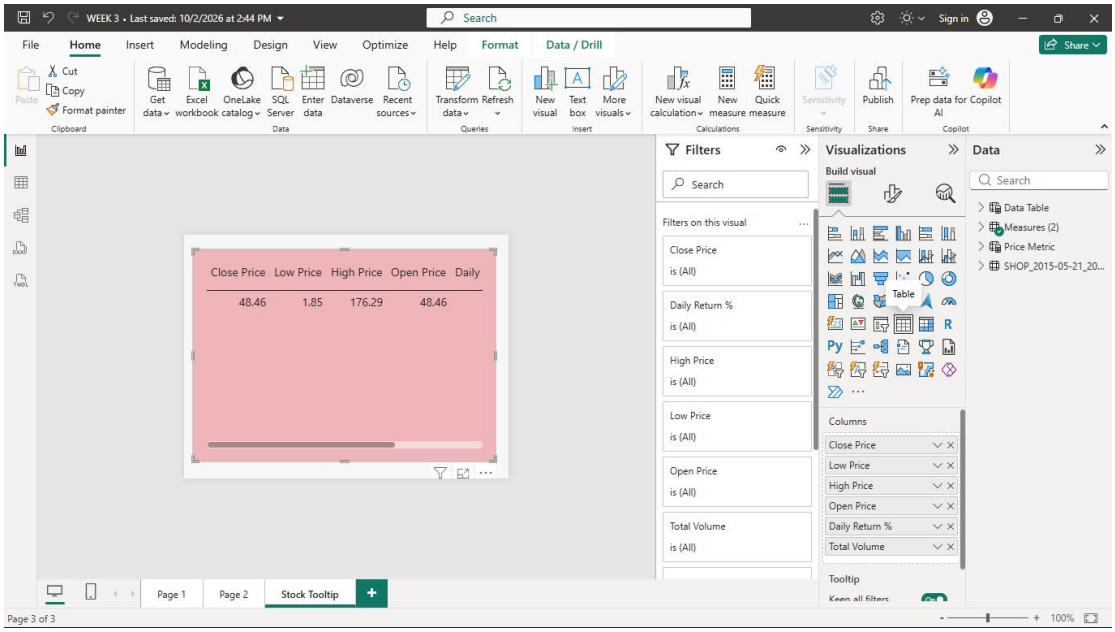
- Open Price
- High Price
- Low Price
- Close Price

Power BI creates a field parameter that can be placed in a slicer.
The user can then select the required price metric and dynamically change the visualization.

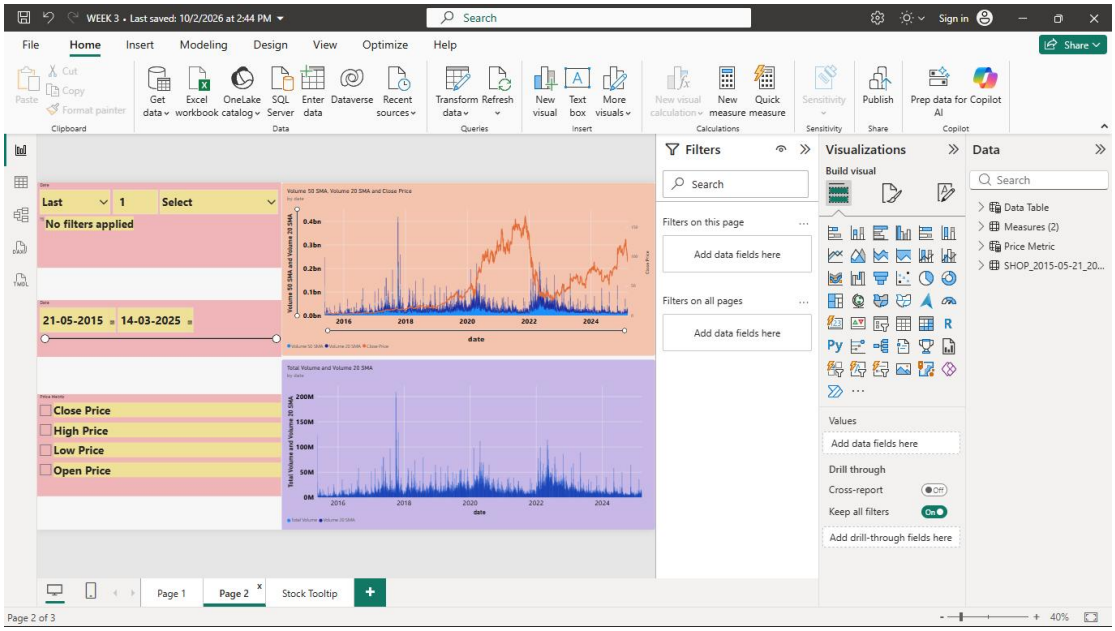


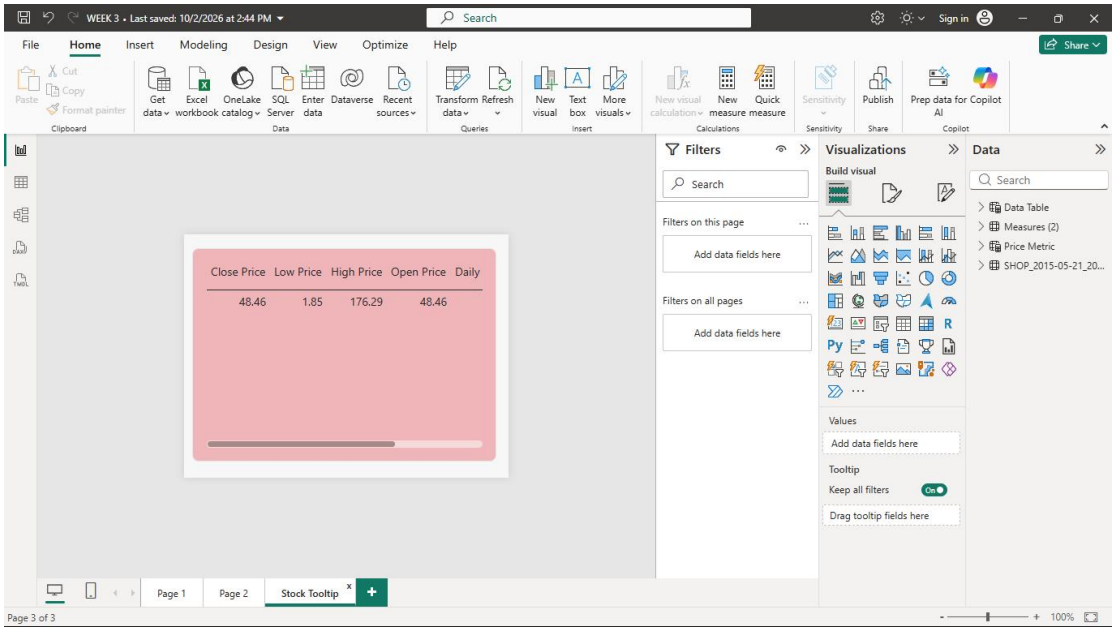
5.)Custom Tooltip Page:

Desing a custom Stock Tooltip page to display Open, High, Low, Close, Volume and Daily Return %.



Final Dashboard:





Conclusion:

The Shopify Financial Performance Dashboard uses Power BI, DAX measures, and interactive visualizations to analyze historical Shopify stock prices and trading volume. The moving averages help users understand short-term and long-term trends, while volume analysis

highlights changes in market activity. Interactive slicers, metric selection, and custom tooltips make the dashboard easier to explore and understand. Overall, the project demonstrates how Power BI can be used to analyze financial data and present meaningful insights through an interactive dashboard.